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The Swedish CArdioPulmonary bioImage Study re-examination: Rationale, design, methods, and management of incidental findings

Elin Good[1,2], Göran Bergström[3,4], Anders Blomberg[5], Viiu Blöndal[6], Gunnar Engström[7,8], Erika Fagman[9], Klas Gränsbo[7,8], Shabab Hasan[10], Shadi Jalali[9], Tomas Jernberg[11], Åse Johnsson[9], Ioannis Katsoularis[5], Jeanette Kuhl[11], Andrei Malinovschi[5], Lina Malm[5], Vibeke Sparring[12], Eva Swahn[1,2], Richard Ssegonja[10], Jelmer Westra[1,2], Mischa Woisetschläger[1,13], Carl Johan Östgren[1] & Emil Hagström[10]

From the [1]Department of Health, Medicine & Caring Sciences, Linköping University, Linköping, Sweden; [2]Department of Cardiology, Linköping University Hospital, Region Östergötland, Linköping, Sweden; [3]Department of Molecular and Clinical Medicine, Sahlgrenska Academy, University of Gothenburg, Gothenburg, Sweden; [4]Department of Cardiology, Sahlgrenska University Hospital, Gothenburg, Sweden; [5]Department of Public Health and Clinical Medicine, Umeå University, Umeå, Sweden; [6]Department of Medical Sciences, Uppsala University, Uppsala, Sweden; [7]Department of Clinical Sciences Malmö, Lund University, Malmö, Sweden; [8]Skåne University Hospital, Malmö, Sweden; [9]Department of Radiology, Institute of Clinical Sciences, Sahlgrenska Academy, Sahlgrenska University Hospital, University of Gothenburg | Region Västra Götaland, Gothenburg, Sweden; [10]Uppsala Clinical Research Centre, Uppsala University, Uppsala, Sweden; [11]Department of Clinical Sciences, Danderyd Hospital, Karolinska Institutet and Karolinska University Hospital, Stockholm, Sweden; [12]Swedish Heart—Lung Foundation, Stockholm, Sweden; and [13]Department of Radiology, Linköping University Hospital, Region Östergötland, Linköping, Sweden

Abstract. Good E, Bergström G, Blomberg A, Blöndal V, Engström G, Fagman E, et al. The Swedish CArdioPulmonary bioImage Study re-examination: Rationale, design, methods, and management of incidental findings. *J Intern Med.* 2026;**299**:467–80.

Objectives. To describe the rationale, design and data collection procedures of the Swedish CArdioPulmonary bioImage Study (SCAPIS) re-examination, which, in its further scope, aims to quantify and explain the development of atherosclerosis, pathological cardiovascular ageing, longitudinal decline in lung function and the malignant transformation of pulmonary nodules among middle-aged Swedes in the longitudinal SCAPIS.

Methods. SCAPIS re-examination is a prospective observational study reassessing approximately 15,000 participants (50% of the original SCAPIS cohort) from six university hospitals. Participants were aged 55–75 years at follow-up, occurring a median of 8.1 years after the baseline investigation. Standardized protocols replicated baseline imaging and functional assessments, including questionnaires, clinical assessments and extensive computer tomography imaging.

Results. Interim analyses of the first 5000 participants (50% women; median age 65.5 [61.8–69.1] years) indicated an expected age-related increase in the prevalence and treatment of hypertension (from 22% to 37%) and diabetes (from 4% to 8%), together with a modest rise in central adiposity. Body mass index (median 26.6 kg/m^2) and the proportion of obesity (22%) remained largely stable, whereas current smoking decreased from 7.5% to 3.4%. The observed patterns were consistent in men and women.

Conclusion. Here we present the rationale, design, methods and management of incidental findings in the SCAPIS re-examination. By integrating serial imaging, functional testing and biomarker profiling, the re-examination will furnish unprecedented insight into cardiopulmonary disease dynamics in an ageing population. These data will underpin personalized risk prediction and inform preventive strategies, while serving as a benchmark for future population-based imaging cohorts.

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Keywords: atherosclerosis, longitudinal studies, preventive medicine

Abbreviations: ACC, American College of Cardiology; ACR, American College of Radiology; AIRC, AI-Rad Companion; ATS, American Thoracic Society; BMI, body mass index; CACS, coronary artery calcium score; CAD, coronary artery disease; CAD-RADS, Coronary Artery Disease–Reporting and Data System; CAL, chronic airflow limitation; CCTA, coronary computed tomography angiography; COPD, chronic obstructive pulmonary disease; CT, computed tomography; DLCO, diffusing capacity of the lung for carbon monoxide; DLCO-SB, single-breath diffusing capacity of the lung for carbon monoxide; ECG, electrocardiography; eCRF, electronic case report form; ERS, European Respiratory Society; FEV1, forced expiratory volume in 1 s; FVC, forced vital capacity; HRCT, high-resolution computed tomography; ILA, interstitial lung abnormality; LDL-C, low-density lipoprotein cholesterol; MIP, maximum intensity projection; PCCT, photon-counting computed tomography; SCAPIS, Swedish CArdioPulmonary bioImage Study; SCORE2, Systematic Coronary Risk Evaluation 2; SCORE2–OP, Systematic Coronary Risk Evaluation 2—older persons; SIS, segment involvement score

Background

Ischaemic heart disease remains the leading cause of death worldwide, accounting for approximately 16% of all deaths [1, 2]. In Western societies, lifestyle changes in the populations and improved primary prevention have addressed some of the major cardiovascular risk factors, such as reductions in tobacco smoking and cholesterol concentrations, contributing to a decline in the incidence of myocardial infarction, together with improved management of acute cardiac care [3–5]. Nevertheless, coronary artery atherosclerosis continues to develop insidiously, often beginning early in life, particularly among men, and remains a major contributor to premature morbidity and mortality [6].

Cardiovascular disease-related mortality is closely followed by chronic respiratory diseases, including chronic obstructive pulmonary disease (COPD) [2]. The global prevalence of COPD continues to rise [7]; the prevalence in Sweden is approximately 7%, with half of the cases accounting for moderate-to-severe disease [8]. COPD is characterized by chronic airflow limitation (CAL) and respiratory symptoms, resulting from prolonged exposure to noxious particles or gases. Many individuals live with CAL for years before being diagnosed, which typically occurs at a late stage when ventilatory capacity is already greatly compromised. Although most COPD patients are current or former smokers, an estimated 20%–25% of individuals with CAL report never having smoked [9, 10].

Despite notable advances in preventive strategies and therapeutic interventions, existing risk-prediction models often fall short in accurately capturing disease progression over time. In this context, large-scale, longitudinal and population-based studies play a pivotal role in advancing our understanding of ageing, subclinical disease development and clinical outcomes. Such studies also provide critical infrastructure for exploratory modelling, biomarker discovery and translational research.

The Swedish CArdioPulmonary bioImage Study (SCAPIS) represents the largest cohort to date using computed tomography (CT) for cardiopulmonary imaging. It was established as a nationwide Swedish initiative to refine disease phenotyping through the integration of advanced imaging modalities, comprehensive biomarker profiling and detailed clinical characterization. SCAPIS was initiated with a pilot study comprising 1111 participants in 2012, which informed the design of the subsequent national data collection phase between 2013 and 2018. In SCAPIS, a random sample of 30,000 men and women aged 50–65 years was recruited from the population at six university sites across Sweden and underwent extensive examinations [11].

SCAPIS re-examination builds upon this foundation and re-examines a random sample of approximately half of the original SCAPIS cohort, thereby enabling longitudinal analyses of disease progression. Leveraging state-of-the-art imaging technologies, serial biomarker measurements and follow-up clinical data, SCAPIS re-examination aims to improve our understanding of cardiopulmonary disease trajectories and enhance the precision

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of future risk prediction models with the goal of reducing the risk of cardiovascular and pulmonary diseases in future generations. We herein describe the rationale, study design, data-collection methods and the framework for the management of incidental findings in SCAPIS re-examination.

Rationale for SCAPIS re-examination

The initial SCAPIS study provided a comprehensive cardiopulmonary assessment through multimodal data collection, including CT, electrocardiography (ECG), spirometry, accelerometery, and extensive biochemical profiling. This rich dataset has enabled novel insights into the early stages and cross-sectional characteristics of cardiopulmonary disease [6, 12–14]. However, to understand the temporal dynamics and natural history of these conditions, longitudinal data are essential. SCAPIS re-examination addresses this need by re-examining approximately half of the original SCAPIS cohort. The repeated measurement of cardiovascular, pulmonary and metabolic parameters has enabled the assessment of long-term changes in cardiometabolic risk and subclinical atherosclerosis, as well as lung function decline, risk of malignancy transformation of pulmonary nodules, and body composition analysis.

The primary objective of the current manuscript is to describe the rationale, design and data collection procedures of the SCAPIS re-examination. In its broader future scope, the SCAPIS re-examination aims to determine characteristics related to the development of atherosclerosis over time, pathological cardiovascular ageing, decline in lung function and malignancy transformation of pulmonary nodules.

Materials and methods

Study population

During the SCAPIS baseline examination (2013–2018), a random sample of 30,000 men and women aged 50–65 years were recruited from the population register at six university hospital sites across Sweden (Gothenburg, Linköping, Malmö/Lund, Stockholm, Umeå and Uppsala) [11]. The recruitment process ensured that the study population was representative of the Swedish population in general [15].

SCAPIS re-examination is a prospective, observational cohort study conducted between 2024 and 2026, designed to longitudinally follow 50% (approximately 15,000) of the individuals from the original SCAPIS cohort (Fig. 1). Individuals who had died between the baseline and the re-examination remain part of the overall SCAPIS study cohort, and their baseline data are included in ongoing and forthcoming outcome analyses on the basis of national register linkages.

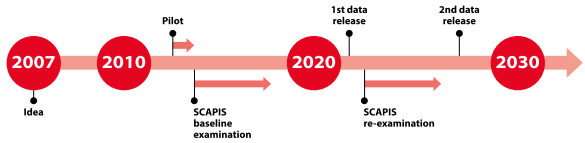


Fig. 1 *Study life span. Schematic timeline of Swedish CArdioPulmonary bioImage Study (SCAPIS) from inception through pilot, baseline examination, re-examination and successive data releases. Arrows indicate examination periods. SCAPIS data are centrally curated with governed access; approved researchers may apply via the SCAPIS database, which is planned to remain available for long-term research use.*

Compared with the SCAPIS baseline investigation, the re-examination employs an almost identical core protocol. The principal differences are that carotid ultrasound, included at baseline, is not part of the re-examination, and that photon-counting CT has replaced the conventional CT systems previously used. Sub-studies led by local investigators varied between the two examination waves: Some were conducted at both time points, whereas others were specific to either the baseline or the re-examination.

Participants in SCAPIS re-examination were randomly selected from those who completed the baseline examination, each site recruiting half of the participants enrolled in the baseline examination. The age range of the participants at the time of re-examination was 55–75 years. The target sample size was determined on the basis of power calculations to ensure sufficient statistical power for key research objectives across cardiovascular and pulmonary domains (see Table S1).

SCAPIS was approved by the Umeå Ethical Review Board (registration number 2010-228-31 M). All participants provided written informed consent. The present re-examination was approved by the Swedish Ethical Review Authority (registration number 2022-06913-01; 2023-04334-02; 2023-07029-02; 2025-01076-02).

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SCAPIS governance structure

SCAPIS has an established and comprehensive governance structure designed to manage and further develop both the baseline study and the subsequent re-examination. The SCAPIS framework, including the pilot, the baseline examination and the re-examination, has been governed by the National Steering Group, and day-to-day operations for the re-examination study were managed by the National Study Lead group. Administrative operations were centralized at the SCAPIS Office, hosted by the University of Gothenburg. The organizational structure also included scientific and operational advisory functions and was designed for continuous development to support long-term research value. Each site was overseen by a local principal investigator, who held overall responsibility for radiation safety, logistical coordination and compliance with ethical standards. An overview of the organizational components and their respective responsibilities is provided in Table S2.

Participant recruitment

Participants were randomly sampled from the SCAPIS baseline cohort. No exclusion criteria were applied, except the inability to understand written and spoken Swedish for informed consent. Invitations were sent by post with study information and a link to an online portal for enrolment confirmation, and those who had relocated were invited to their original study centre. Invitees who registered interest by submitting contact details were contacted by telephone up to three times to schedule the visit. A letter with a reminder about the study was also sent to all participants who had not visited the SCAPIS web page approximately 1 month after the initial invitation. If the centre was contacted and the participant was willing to participate in the study, an appointment was arranged at the study centre. To date, the participation rate has remained stable at about 76%. No reimbursement was provided for travel expenses or loss of income. Fig. 2 displays a flow chart detailing the inclusion of participants in the study.

Data collection

Data collection was standardized across all six study centres, ensuring consistency with methodologies in the baseline SCAPIS data collection. In accordance with the protocol, core examinations at the study centre and the CT were scheduled to occur within approximately 14 days after consent;

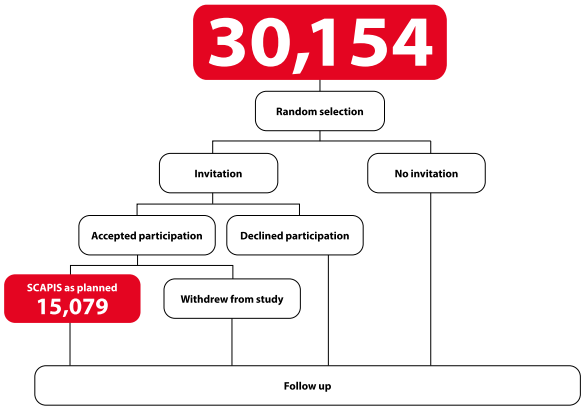


Fig. 2 *Inclusion flow chart. Overview of the randomized selection of participants for the Swedish CArdioPulmonary bioImage Study (SCAPIS) re-examination, based on the original 30,154 participants in the SCAPIS baseline study. Approximately 50% of the baseline cohort (target N = 15,089) will be invited to the re-examination. Data collection is ongoing, and final inclusion numbers and reasons for nonparticipation will be available after study completion in 2026. Compared with the baseline protocol, the re-examination does not include carotid ultrasound but otherwise replicates the core assessments, with passive follow-up through Swedish national health registers.*

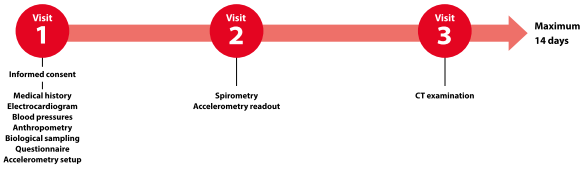


Fig. 3 *General description of the visits. The figure represents the general visit scheme, disregarding some minor differences between sites regarding visits 1 and 2. In addition to the Swedish CArdioPulmonary bioImage Study (SCAPIS) re-examination examinations, 81 local substudies were performed with additional examinations during the visits or as separate visits. CT, computed tomography.*

actual scheduling intervals varied with operational capacity. Fig. 3 presents a general description of what each visit entailed.

Participants underwent a comprehensive assessment protocol, including the following points:

- **Questionnaires**: Detailed information on diet, medications, current and prior diseases, environmental and lifestyle factors, psychosocial well-being, socioeconomic status and other social determinants was collected through an

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on-site questionnaire comprising 48 main items with sub-questions. Participants reported current and past diseases, allergies and current medication (Table S3).

- **Anthropometry and blood pressure**: Weight, height and waist-to-hip ratio were registered. Brachial arterial blood pressure, ankle blood pressure and ankle brachial index were measured (Table S4). An Omron M7 Intelli IT-AFIB and blood pressure cuff were used for brachial measurements. Systolic and diastolic pressures were registered in supine position after 5 min of quiet rest, and the mean of two stable readings was documented. For ankle brachial index, a Hadeco Bidop ES-100V3 Doppler and a manual blood pressure cuff were used.
- **Biochemical analyses**: Blood samples (whole blood, plasma and serum) were drawn at the first visit from each participant, who had fasted overnight (at least 8 h) and was encouraged not to smoke. Plasma levels of haemoglobin, blood platelets, differential leukocyte count, C-reactive protein, creatinine, glucose, glycated haemoglobin, triglycerides, total cholesterol, low-density lipoprotein cholesterol (LDL-C) and high-density lipoprotein cholesterol were analysed. The biochemical analyses were analysed at the local university hospital laboratory (Table S5 for details). For participants with elevated plasma glucose levels, a repeated overnight fasting glucose measurement was performed. Venous blood and spot-urine for biobanking were processed and stored at the local Biobank facilities in collaboration with Biobank Sverige (biobanksverige.se). To ensure complete traceability of samples and related information, all codes and pre-analytical steps were controlled by the Biobank facilities Laboratory Information Management System. Time from sampling to freezer was recorded, with the aim to freeze all samples within 2 h after sampling.
- **ECG**: Standard 12-lead ECG equipment was used with the possibility of storing data (raw-ECGs) electronically. Output was 50 mm/s.
- **Accelerometry**: The wearable device ActiGraph Activity Monitor GT3X-BT accelerometer (ActiGraph) was used to measure physical activity levels. The accelerometer is worn for 7 days on the right side of the hip, attached to an elastic belt (see Table S6 for details).
- **Lung function tests**: Participants were recommended to avoid smoking on the same day as the lung functions examinations. Vyntus ONE DL (Jaeger Medical) was used at all sites to perform spirometry and DLCO measurements. All sites used the same software version, SentrySuite 3.20.8. A dynamic spirometry was performed to assess lung function at least 15 min after the inhalation of a short-acting beta-2-agonist (400 μg salbutamol administered via spacer in four doses of 100 μg each). Diffusion capacity of the lungs for carbon monoxide, also referred to as the transfer factor of the lungs for carbon monoxide, was assessed using the single-breath approach (DLCO-SB). Participants were instructed to exhale completely, then inhale the gas mixture maximally, hold their breath for at least 10 s and then exhale.

All pulmonary function testing was performed in accordance with a standardized operating procedure aligned with the most recent guidelines from the American Thoracic Society and the European Respiratory Society [16, 17]. To ensure adherence to these standards and consistency across study sites, on-site visits from study coordinators were conducted at participating centres. Impulse oscillometry was performed as an add-on study in Uppsala and Malmö. Unlike other centres, Uppsala and Umeå conducted pulmonary function assessments both before and after bronchial dilatation:

- **CT imaging**: The CT protocol included inspiratory and expiratory chest CT for lung tissue assessment, low-dose body CT for body composition analysis, coronary artery calcium CT and coronary computed tomography angiography (CCTA) for coronary artery evaluation. CT image acquisition and image reconstruction parameters are shown in Table 1. To ensure standardized conditions for the whole-body CT, all participants were instructed to consume a standardized meal before the visit, with the caloric content tailored to the participant's estimated daily energy expenditure.

All imaging was performed using a dual source photon counting detector CT (NAEOTOM Alpha, Siemens Healthineers). Pulmonary imaging was performed without contrast media. To enhance image quality (IQ) for the coronary assessment, the beta-blocker metoprolol could be administered

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Table 1. *Computed tomography (CT) acquisition parameters across SCAPIS 2 imaging domains.*

Parameter	CaScore (flash)	CaScore (sequence)	Thorax (inspiration)	Thorax (exspiration)	Liver/ abdomen/ thigh slices	Whole body
Scan mode	Cardiac flash spiral adult quantum plus	Cardiac dual source sequence adult quantum plus	Routine flash spiral adult high res quantum plus			
Rotation time (s)	0.25	0.25	0.25	0.25	0.25	0.25
Pitch	3.20	–	2.20	2.20	1.6	1.6
Image quality level	19	19	40	10	5	5
Tube voltage (kV)	120/140				140	140
CARE keV optimization	Non-contrast					
Scan phase (%)	65	75–75	–	–	–	–
Acquisition (mm)	144 × 0.4	132 × 0.4	96 × 0.2	96 × 0.2	96 × 0.2	96 × 0.2

Note: Parameters for coronary calcium scoring (CaScore), chest scans during inspiration and expiration, and body composition imaging (liver, abdomen, thigh and whole-body) are shown. All scans were performed using high-resolution protocols optimized for non-contrast imaging.
Abbreviations: CARE, combined applications to reduce exposure; CaScore, coronary artery calcium score; IQ, image quality; keV, kilo-electron Volt; kV, kilovolt; mm, millimetre; s, seconds.

to reduce and stabilize heart rate, aiming for a heart rate below 60 beats/min. Metoprolol was administered orally 1.5–2 h before the examination (up to 100 mg) and/or intravenously immediately before the examination (up to 15 mg), together with sublingual nitroglycerin. Contrast medium (Omnipaque 350 mg I/mL) was administered, unless contraindicated, at a dose of 244–281 mg/kg body weight depending on the CCTA scan mode. Additional methodological details on CT imaging, including quality assurance procedures involving comparative dual imaging using both photon-counting and conventional CT technology, are provided in the Methods section of Supporting Information, and a detailed summary of the findings and corresponding follow-up procedures is provided in Table S7.

In contrast to the SCAPIS baseline protocol, a low-dose thoracoabdominal CT scan (approximately 1 mSv in total) was additionally acquired at three study sites: Gothenburg, Linköping and Uppsala. This scan extended from the thoracic inlet to just below the mid-thigh, supplementing the originally acquired three anatomical regions. At the remaining three study sites, imaging was limited to the same three regions as in the baseline assessment: a single CT slice through the liver, one through the L4 vertebral level and one through the mid-femur.

In addition to the assessments included in the core study, each site conducted several local substudies, many of which involved additional measurements and data collection. In total, 81 local substudies were carried out, either integrated into the main study visits or as separate visits, provided in Table S8.

Data management and secure transfer procedures

In SCAPIS re-examination, data handling and transfer were carried out according to prespecified protocols to ensure high data quality, integrity and compliance with legal and ethical standards. Detailed procedures for data management, quality assurance and quality control are described in the Methods section in the Supporting Information. Most clinical and questionnaire data were recorded directly into electronic case report forms (eCRF). Each participant kept the same unique study ID as in the baseline assessment. Additional raw data, such as CT images, spirometry flow curves, ECG traces, accelerometery outputs and laboratory results, were securely transferred from each hospital to the University of Gothenburg, using encrypted communication channels to prevent data corruption or unauthorized access. The frequency of data transfer varied by dataset and site: CT images were typically transferred daily, spirometry data quarterly, ECG data monthly, accelerometery data weekly and laboratory data monthly. All data were stored within the University of Gothenburg's secure data environment, maintained

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by its IT department and compliant with high standards of physical and digital security.

Hospital Biobanks were allowed to store biological samples linked to participants' personal identification numbers to ensure traceability, in accordance with the Swedish Biobanks in Medical Care Act (SFS 2023:38).

Data assessment

Cardiovascular imaging. CT images were analysed using the software *syngo.via* VB80B (Siemens Healthineers). Following image reconstruction, the calcium content in each artery was automatically quantified using the *syngo.via* CT CaScoring software, on the basis of the calcium scoring method described by Agatston et al. [18]. Manual corrections of the automated scoring were performed by the reader when necessary. CCTA findings were reported using the 18-segment coronary artery model [19]. Each segment was categorized as having no atherosclerosis, 1%–49% stenosis, or ≥50% stenosis (i.e., significant stenosis). Additionally, the presence of calcified and non-calcified plaques, as well as non-diagnostic segments due to calcium blooming or technical limitations, were assessed on a per-segment level. The maximum degree of stenosis was reported according to the Coronary Artery Disease–Reporting and Data System (CAD-RADS) 2.0 classification, and the presence of high-risk features of atherosclerosis (plaques) was documented [20].

Total Agatston score per participant was categorized according to CAD-RADS 2.0, with the highest two categories (severe and extensive) combined, yielding the following groups: 0, 1–100, 101–300 and >300 Agatston units [20]. CCTA findings were further summarized by segment involvement score (SIS; number of segments with atherosclerosis) using CAD-RADS 2.0 categories: 0, 1–2, 3–4 and ≥5; the presence of obstructive disease (≥50% stenosis in any segment) and the presence of non-calcified atherosclerosis.

Pulmonary imaging. Airway abnormalities, emphysema, interstitial lung changes, pulmonary nodules and other findings of potential concern—including suspected malignancies—were systematically evaluated using a modified score sheet inspired by COPD Gene study [21] and documented in the eCRF (Table S9; lung nodules, Table S7).

A stepwise workflow was assigned in *syngo.via* to enable a thorough and standardized evaluation of the lung parenchyma. AI-Rad Companion (AIRC tool), Siemens Healthineers, was used for quantitative analyses of emphysema, pulmonary nodules and dimensions of the thoracic aorta. AIRC assessments of pulmonary nodules and aortic dimensions were presented to the radiologists, but the results from AIRC emphysema scoring were sent directly to the SCAPIS re-examination database for later use in research, and the visual emphysema scoring was performed without knowledge of the AIRC results.

Multiplanar reformats of 0.4 mm slice thickness were used in the evaluation of the lung parenchyma: bronchiectasis; bronchial wall thickening; emphysema; ground glass changes; mosaic attenuation; cysts; reticular abnormalities; honeycombing; post-inflammatory changes and scarring; consolidations; previous changes due to treatment like radiotherapy; and surgery or transplantation and lung collapse (atelectasis).

To increase sensitivity for the detection of emphysema and air trapping, minimum intensity projections of 10 mm images were additionally applied on the inspiratory and expiratory images. Emphysematous changes were visually categorized as mild if involving 1%–25% of the lung parenchyma, moderate if involving >25%–50% and severe if involving >50%–100% of the lungs. It was described as centrilobular, paraseptal or advanced destructive as appropriate. Features of air trapping, as represented by persistent mosaic attenuation, were also reported as a possible indication of small airway disease.

The search for pulmonary nodules was additionally performed with maximum intensity projections of 5 mm and using the AIRC tool as a second reader for nodule detection and size estimation, including the assessment of nodule growth compared with the baseline SCAPIS investigation. Nodule growth was assessed at a slice thickness of 0.8 mm. The AIRC tool flagged all suspicious pulmonary nodules and provided estimates of dimensions and volume. The nodules assessed included solid nodules, part-solid/semisolid nodules, ground glass nodules and perifissural nodules (defined as <10 mm). If the AIRC tool measurements were deemed insufficient or if nodules were missed, the radiologist manually flagged, measured and recategorized them in accordance with Fleischer's guidelines

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[22]. The largest new nodule, fastest growing nodule and largest known nodule for solid, part-solid and ground glass nodules were noted in the eCRF. When assessing the nodules, perifissural nodules were not classified as solid nodules. All other findings suspicious for malignancy that did not meet the criteria for classification as nodules (such as lesions exceeding 3 cm in diameter) were flagged and directly reported to the study physician.

Body composition imaging. Review of the body images was restricted to safety screening and aimed solely at identifying incidental findings rather than providing a systematic diagnostic assessment. Reportable observations were entered in the eCRF under the following categories: adrenal glands, gynaecological structures, hernias, vasculature, lymph nodes, liver, kidneys, pancreas, skeletal abnormalities and other.

Definition and communication of incidental findings

Management and feedback of incidental and pathological findings

SCAPIS re-examination was a population-based research study rather than a health-screening programme; accordingly, only a limited subset of clinically significant findings was communicated, and direct contact with participants was restricted to potentially serious conditions. A detailed description of the structured and participant-centred communication procedures is provided in the Methods section of Supporting Information.

The feedback system for incidental and pathological findings was developed through national consensus within the SCAPIS working group, in collaboration with clinical specialists in cardiology, pulmonology, nephrology, gynaecology and abdominal radiology, and aligned with national and international guidelines. The protocol was based on predefined cut-off values and response pathways for clinically relevant test results.

In Sweden, all results documented in electronic medical records are accessible online to patients and study participants. A guiding principle was therefore to avoid unnecessary concern by withholding benign or stable findings unlikely to require clinical action. All results were first reviewed by a local study physician, who assessed their clinical significance. Complex findings from coronary, pulmonary or abdominal CT were referred to for additional specialist input, either through multidisciplinary team meetings or consultation with hospital sub-specialists. Although local procedures varied slightly across study sites, all physicians had access to these resources, and all decisions were documented in the medical record.

Incidental findings of uncertain character prompted further imaging and follow-up, and new diseases requiring treatment led to referral to the appropriate healthcare provider according to predefined decision support. Designated study physicians ensured that no clinically relevant findings requiring action were overlooked. Once a participant entered clinical care because of a pathological finding, responsibility for further diagnostics and treatment was transferred to the regional healthcare system.

Management of expected and incidental findings

Decision support: follow-up of findings

To ensure standardized clinical management and participant safety, SCAPIS re-examination incorporated a structured decision-support framework for handling incidental and pathological findings on the basis of national and international guidelines. Predefined criteria guided the clinical interpretation and management of radiological abnormalities, ensuring consistency across study sites. Radiologists and study physicians evaluated findings using standardized protocols, with referral decisions based on severity, medical history and symptomatology. Table S7 details the integrated decision support structure applied to radiological findings.

Clinically relevant results were reviewed by designated study physicians, who determined appropriate actions aligned with predefined criteria developed in collaboration with national clinical experts. The general principles guiding responses to clinical measurements and laboratory findings are summarized in Table S10. The framework prioritized an early detection of potentially serious conditions while minimizing unnecessary interventions for benign or non-actionable findings. All suspected malignancies were immediately referred without delay.

Cardiovascular risk assessment

Each participant received a personalized letter from the study physician summarizing their

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cardiovascular risk profile and indicating prevention targets (e.g., blood pressure and LDL-C). Cardiovascular risk was quantified in apparently healthy individuals using the Systematic Coronary Risk Evaluation 2 (SCORE2) or Systematic Coronary Risk Evaluation 2—older persons algorithm, according to age [23, 24]. Participants with diabetes mellitus or a history of atherosclerotic cardiovascular disease were considered as having high or very high cardiovascular risk. Having coronary artery calcium score (CACS) ≥100%, SIS ≥4%, or a ≥50% coronary artery stenosis defined the patient as having established coronary artery disease (CAD) [20, 25]. The SCORE2-derived risk estimates, risk states and imaging-derived findings of CAD, formed the basis for individualized risk stratification [26–29]. The risk classification directed the treatment recommendations according to national and international guidelines, including the LDL-C and blood-pressure targets. Table S11 outlines the LDL-C thresholds and corresponding therapeutic strategies applied in SCAPIS re-examination. Participants with established CAD met a study nurse for a brief, structured discussion of modifiable risk factors and usual care. They were advised to seek their primary care physician for reassessment of the cardiovascular risk, and, if indicated by local routines, initiate preventive treatment. The study team did not prescribe or manage therapy. After 3 months, the participant was contacted by telephone to reassure that a primary care contact had been established. Patients with a history of atherosclerotic cardiovascular disease or diabetes mellitus were not assessed at the study site, based on the assumption that they already had ongoing healthcare contacts and had received optimized preventive care.

Pulmonary assessment

The pulmonary assessment and management of incidental findings were guided by national and international guidelines, and—in the absence of such guidelines—by study-specific protocols developed for both SCAPIS baseline and SCAPIS re-examination.

Forced expiratory volume in 1 s (FEV1) was used as an indicator of CAL in the spirometry assessment and was tested 15 min after the administration of bronchodilation. Per cent–predicted values for FEV1, forced vital capacity (FVC), and DLCO were calculated using the European reference equations for spirometry and DLCO [30, 31]. FEV1 estimates between 50% and 80% of predicted, and FVC estimates between 50% and 70% without previous lung disease, led to a referral to the participant's primary healthcare physician for further evaluation. All FEV1 and/or FVC values below 50% of predicted prompted referral to specialist care, as such reductions reflect significant loss of lung function. Participants with a DLCO value below 60% of the predicted level were referred to the lung clinic for further evaluation. These thresholds are in line with published recommendations for spirometry-based triage in primary care settings [32]. Participants with reduced lung function were not referred if the findings were attributable to a previously diagnosed lung disease or if they were already under active management.

A direct comparison with the corresponding CT scan from the baseline SCAPIS investigation was performed to assess IQ and interval changes. Pulmonary nodules (<3 cm) were managed in accordance with national lung cancer care programme guidelines [23] and the Fleischer Society recommendations. Risk stratification and management decisions were guided by the participant's risk profile, primarily prior malignancy and radiological features such as growth on serial imaging, spiculated margins, multiplicity and upper lobe location. As a general rule, nodules with a volume doubling time >600 days, or those that remained stable in size and morphology, were not subject to further follow-up.

Follow-up intervals were generally adapted from Swedish national guidelines and based on size, number and morphology of the nodule (Table S7). Individualized follow-up was applied when appropriate, following consultation with a thoracic radiologist and/or lung specialist.

Suspected lung malignancies were referred without delay to a lung clinic for evaluation within a standardized cancer care pathway. This process includes specialist review, histological confirmation when applicable and a treatment decision within 4–6 weeks—a framework shown to promote timely cancer diagnosis [33].

Interstitial lung abnormalities, such as honeycombing, reticulation or traction bronchiectasis affecting more than 5% of the lung parenchyma, triggered referral to a lung specialist. Similarly, widespread cystic lung disease prompted specialist evaluation. In contrast, emphysematous changes,

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Table 2. *Overview of SCAPIS cohort characteristics across study phases.*

	SCAPIS pilot	SCAPIS baseline	SCAPIS re-examination	Change
Characteristics				
Year of examination	2012	2013–2018	2024–2026	–
Number	1111	30,154	5000	–
Age, years	57.6 (53.8–61.7)	57.4 (53.7–61.2)	65.5 (61.8–69.1)	+8.1 years
Age span, years	50–65	50–65	55–75	–
Sex, female	50	50	50	0
Participation rate	49.5	50.3	≈76[a]	
Disease prevalence				
Hypertension	36.0 (400)	21.9 (6618)	36.8 (1838)	+14.9%
Antihypertensive treatment	23.1 (257)	19.1 (5766)	34.6 (1729)	+15.5%
Diabetes mellitus	6.8 (1024)	4.3 (1291)	8.4 (418)	+4.1%
Treatment for diabetes	5.6 (62)	3.6 (1086)	7.3 (367)	+3.7%
Cancer	7.4 (82)	5.8 (1738)	10.9 (546)	+5.1%
Risk factors				
BMI (kg/m^2)	26.6 (24.3–29.5)	26.3 (23.9–29.4)	26.6 (24.0–29.5)	0.3 kg/m^2, +1.1%
Obesity (BMI ≥ 30 kg/m^2)	22.1 (245)	21.5 (6483)	22.3 (1115)	+0.8%
Waist circumference, cm	96 (86–103)	94 (85–103)	96 (88–105)	+2 cm, +2.1%
Smoking				
Never	42.8 (476)	48.8 (14,729)	51.7 (2584)[b]	+2.9%
Former	38.6 (429)	35.2 (10,601)	41.9 (2097)	+6.7%
Occasional smoker	3.3 (37)	4.8 (1437)	2.3 (114)	−2.5%
Current	14.7 (163)	7.5 (2250)	3.4 (171)	−4.1%

Note: Demographic characteristics, participation rates, disease prevalence, and key cardiometabolic risk factors are presented for the SCAPIS pilot, baseline and re-examination phases. The table includes data for the first 5000 participants of SCAPIS re-examination and indicates change from baseline. Where applicable, values are presented as percentages with absolute counts in parentheses or as medians with interquartile ranges. Data are median (interquartile range) and % (number).

Abbreviations: BMI, body mass index; cm, centimetres; kg/m^2, kilograms per square meter; SCAPIS, Swedish CArdioPulmonary bioImage Study.

[a]Re-examination participation rate is the proportion of invited eligible participants who completed a SCAPIS re-examination visit; the baseline cohort size is not used as the denominator.

[b]Re-examination values are interim estimates based on the first 5000 participants with available data; figures will be updated as inclusion progresses.

symptomatic bronchiectasis and findings indicative of inflammation or infection were referred to primary care, with specialist support when needed. A full overview of the radiological lung assessment protocol is provided in Table S9.

Preliminary results

Among the first 5000 SCAPIS re-examination participants, median age increased from 57.4 at baseline to 65.5 years at re-examination, and the sex distribution remained balanced at 50% female. Table 2 summarizes cohort characteristics across phases.

From baseline to re-examination in the interim analysis, the cardiometabolic risk burden increased: Hypertension and diabetes were more prevalent and pharmacological treatment for these diseases were more common; central adiposity rose slightly, whereas body mass index and proportion of obesity remained largely stable. Although current smoking declined, this favourable trend appears insufficient to offset the overall worsening

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cardiometabolic profile. Estimates will be refined as data accrual proceeds.

Discussion

The SCAPIS re-examination study represents an opportunity to study the long-term development of cardiovascular and pulmonary diseases in a meticulously characterized cohort from a middle-aged Swedish population. In this manuscript, we present the data collection procedures underlying the re-examination. The longitudinal design, together with comprehensive phenotypic characterization, offers a unique platform for advancing our understanding of disease trajectories, with the ultimate goal of developing personalized preventive strategies and public health policies based on evidence. By re-examining 15,000 participants 8–10 years after the initial assessment in the SCAPIS baseline investigation, SCAPIS re-examination will deliver unmatched disease progression insights, potential interventions, and emerging clinical markers relevant to cardiovascular and pulmonary medicine.

In comparison to international research initiatives, the combined SCAPIS baseline and re-examination phases have an important and unique niche. The Framingham Heart Study, a classic longitudinal study that began in 1948 and has contributed much to our current knowledge about cardiovascular disease, is not as imaging-intensive as SCAPIS, but it has a strong focus on cardiovascular risk. The UK Biobank, similar in scope but with broader genetic and lifestyle data across 500,000 participants, does not provide imaging detail equal to SCAPIS. The Multi-Ethnic Study of Atherosclerosis [34] is focused on imaging but includes a smaller number of participants ($N = 6500$) and includes assessment of CACS but not CTTA or lung tissue assessments. Similarly, the German National Cohort (NAKO Gesundheitsstudie) shares the comprehensive approach of SCAPIS but covers a larger population ($N = 30{,}861$) without the specific depth in cardiopulmonary imaging [35]. Thus, SCAPIS uniquely integrates advanced cardiopulmonary imaging at two time points with extensive longitudinal health data from Swedish national registers. To date, SCAPIS has generated 251 peer-reviewed publications, illustrating substantial scientific impact across multiple disciplines.

The SCAPIS baseline examination, together with the SCAPIS re-examination's longitudinal design, now enables causal inference through time-sequenced data, opening new possibilities for modelling risk factor trajectories, assessing cumulative exposures and identifying early predictors of clinical outcomes. The cohort also provides material for state-of-the-art research using artificial intelligence and machine learning in imaging analysis, multiomics integration and register-based health economics.

Using the unique personal identification number assigned to all Swedish citizens, data from SCAPIS baseline and SCAPIS re-examination will be linked to national health registries and other data sources, including the Cause of Death Register, the Swedish Prescribed Drug Register and the National Patient Register, which contains information on hospital admissions and outpatient specialist visits. This linkage will enable the association of findings from SCAPIS baseline and SCAPIS re-examination with longitudinal outcomes such as prognosis, pharmacotherapy and other relevant health data.

An important ethical and practical challenge in population-based imaging studies is managing incidental findings—unintentionally detected abnormalities that become more common with age—such as pulmonary nodules, coronary artery stenosis, aortic aneurysms, renal cancer and vertebral fractures. Although incidental findings can facilitate early detection of clinically significant conditions, they may also cause psychological distress, overtreatment and unnecessary healthcare use. SCAPIS addresses these complexities through standardized protocols for ethically sound identification, communication and clinical management. However, despite these measures, there remains an increased healthcare burden, at least in the short term. Participant safety was systematically monitored. Study physicians managed incidental findings and related clinical referrals. Serious incidents were documented, triggering corrective actions such as retraining or intensified monitoring, and reviewed by a structured governance process involving local coordinators, national leadership and the SCAPIS steering group.

SCAPIS strongly advocates open science and data transparency. Data and biospecimen access are managed through a structured application process (detailed at www.scapis.org), fostering collaboration and maximizing scientific output. As SCAPIS re-examination progresses, it will serve

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as a national and international model for longitudinal population-based imaging studies, significantly informing clinical practice and public health policy.

Strengths and limitations

One of the key strengths of SCAPIS re-examination lies in its population-based foundation, comprehensive use of advanced imaging technologies, biomarker profiling, and extensive clinical, lifestyle and environmental data. This multidimensional approach enables within-subject analyses of longitudinal changes, allowing for the early detection of subclinical alterations in cardiovascular and pulmonary structure and function. The use of harmonized protocols across all participating university hospital sites ensures consistent data quality, enhancing both comparability and internal validity.

SCAPIS re-examination also faces inherent limitations common to longitudinal cohort studies. Participant attrition may introduce selection bias, especially if associated with poorer health or lower socioeconomic status. The primarily urban recruitment strategy potentially limits generalizability to rural populations. Moreover, the comprehensive study design demands substantial logistical, financial and human resources, potentially constraining scalability and feasibility in other settings.

Conclusion

Building on the extensive data collected during the SCAPIS baseline examination, SCAPIS re-examination seeks to address critical knowledge gaps in the progression of chronic cardiovascular and pulmonary diseases. The design of the study, which includes multimodal imaging, biomarker profiling and comprehensive clinical assessments, is poised to make a significant contribution to cardiovascular and pulmonary research.

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Conflict of interest statement

The authors declare no conflicts of interest.

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Data availability statement

The data underlying this article are available through the SCAPIS data access platform www.scapis.org. Access to derived data generated within the SCAPIS project is granted on a case-by-case basis following review. Applications for data access must be submitted electronically via the SCAPIS platform.

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Correspondence: Elin Good, Linköping Cardiology Clinic, Linköping University Hospital, 581 85 Linköping, Sweden.
Email: elin.good@liu.se

Supporting Information

Additional Supporting Information may be found in the online version of this article:

Table S1: SCAPIS re-examination power calculations.

Table S2: Overview of organizational components and responsibilities.

Table S3: Summary of participant questionnaire.

Table S4: Methodology for anthropometry and blood pressure measurements in SCAPIS.

Table S5: Detailed overview of blood status and biochemical analyses.

Table S6: Accelometry, methodological summary.

Table S7: Decision support: Actions based on radiological findings.

Table S8: Summary of SCAPIS re-examination substudies.

Table S9: Radiological lung assessment.

Table S10: Decision support: Actions based on findings from clinical measurements and laboratory values.

Table S11: LDL-C management.

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